

Complete DevOps Guide: Docker, Kubernetes, Jenkins & CI/CD

1. What is DevOps?

DevOps is a culture and set of practices that combine Development (Dev) and Operations (Ops). It focuses on faster software delivery, automation, continuous integration, continuous deployment, monitoring, and collaboration between teams.

2. DevOps Lifecycle

- Plan
- Develop
- Build
- Test
- Release
- Deploy
- Operate
- Monitor

This lifecycle is continuous and automated.

3. Version Control (Git)

Git is a distributed version control system.

Common commands:

```
git init
```

```
git add .
```

```
git commit -m "message"
```

```
git push
```

Platforms: GitHub, GitLab, Bitbucket.

4. Continuous Integration (CI)

CI means automatically building and testing code whenever developers push changes.

Benefits:

- Early bug detection
- Automated testing
- Faster development

Tools: Jenkins, GitHub Actions, GitLab CI.

5. Continuous Deployment (CD)

CD automatically deploys applications after successful testing.

Types:

- Continuous Delivery (manual approval before deploy)
- Continuous Deployment (fully automatic)

6. Jenkins

Jenkins is an open-source automation server.

Used to build, test, and deploy applications.

Key concepts:

- Pipelines
- Jobs
- Agents
- Plugins

Pipeline example stages:

1. Build
2. Test
3. Deploy

7. Docker

Docker is a containerization platform.

Containers package application + dependencies.

Important concepts:

- Image
- Container
- Dockerfile
- Docker Hub

Common commands:

```
docker build
docker run
docker ps
docker stop
```

8. Dockerfile

Used to create Docker images.

Common instructions:

FROM
WORKDIR
COPY
RUN
EXPOSE
CMD

9. Docker Compose

Used to run multi-container applications.

Example services:

- Backend
- Frontend
- Database

Command:

`docker compose up`

10. Kubernetes

Kubernetes (K8s) is a container orchestration platform.

It manages container deployment, scaling, and networking.

Core components:

- Pod
- Deployment
- Service
- ReplicaSet
- Namespace

11. Kubernetes Architecture

Master Node:

- API Server
- Scheduler
- Controller Manager

Worker Node:

- Kubelet
- Kube Proxy
- Container Runtime

12. Infrastructure as Code (IaC)

IaC means managing infrastructure using code.

Tools:

- Terraform
- Ansible
- CloudFormation

Benefits:

- Automation
- Version control
- Reproducibility

13. Monitoring & Logging

Monitoring tools:

- Prometheus
- Grafana
- ELK Stack

Used to track performance, errors, and system health.

14. Cloud Platforms

Common Cloud Providers:

- AWS
- Azure
- Google Cloud

Services include compute, storage, networking, databases.

15. CI/CD Pipeline Example

1. Developer pushes code to GitHub
2. Jenkins triggers build
3. Tests run automatically
4. Docker image created
5. Image pushed to Docker Hub
6. Kubernetes deploys new version

16. DevOps Best Practices

- Automate everything
- Use version control
- Small frequent releases
- Monitor continuously
- Secure pipelines

17. Security in DevOps (DevSecOps)

Integrate security in pipeline:

- Static Code Analysis
- Dependency Scanning
- Container Scanning
- Secrets Management

18. Real-World DevOps Workflow

Frontend (React) + Backend (Node.js) + Database (MongoDB)

- Step 1: Code pushed to GitHub
- Step 2: Jenkins builds Docker image
- Step 3: Image stored in registry
- Step 4: Kubernetes deploys container
- Step 5: Monitoring tools track health